Second-Order Models in RSM

When the curvature is detected, the experimenter is relatively close to the optimum, so a model that incorporates curvature is usually required to approximate the response.

$$y = \beta_0 + \sum_{i=1}^{k} \beta_i x_i + \sum_{i=1}^{k} \beta_{ii} x_i^2 + \sum_{i<j}\sum \beta_{ij} x_i x_j + \varepsilon$$

- Second-order models are used widely in practice
- The Taylor series analogy
- Optimization is easy
- There is a lot of empirical evidence that they work very well

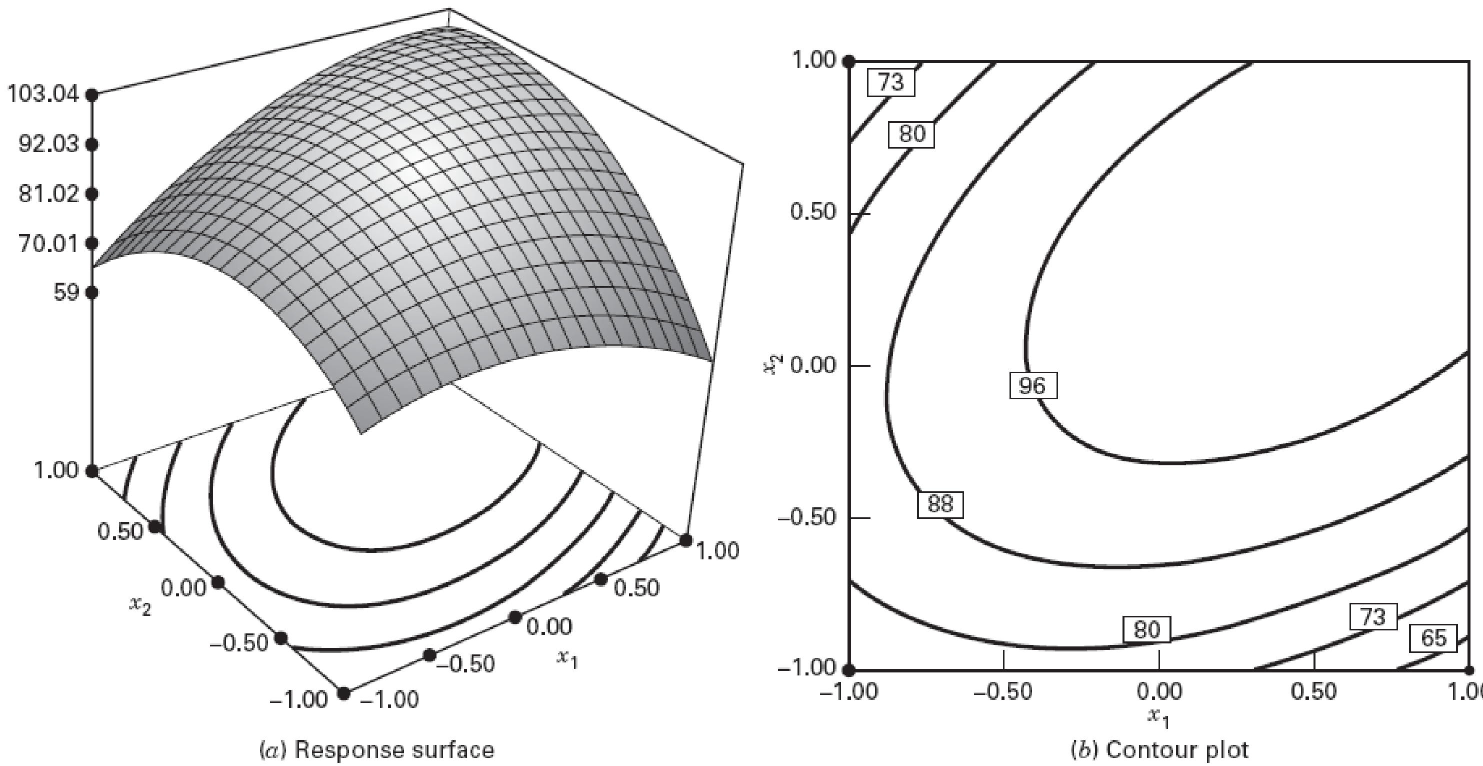
Stationary Point

- Suppose we wish to find the levels of $x_1, x_2, \dots, x_k$ that optimize the predicted response.
- This point, if it exists, may be the set of $x_1, x_2, \dots, x_k$ for which the partial derivatives

$$\partial\hat{y}/\partial x_1 = \partial\hat{y}/\partial x_2 = \cdots = \partial\hat{y}/\partial x_k = 0$$

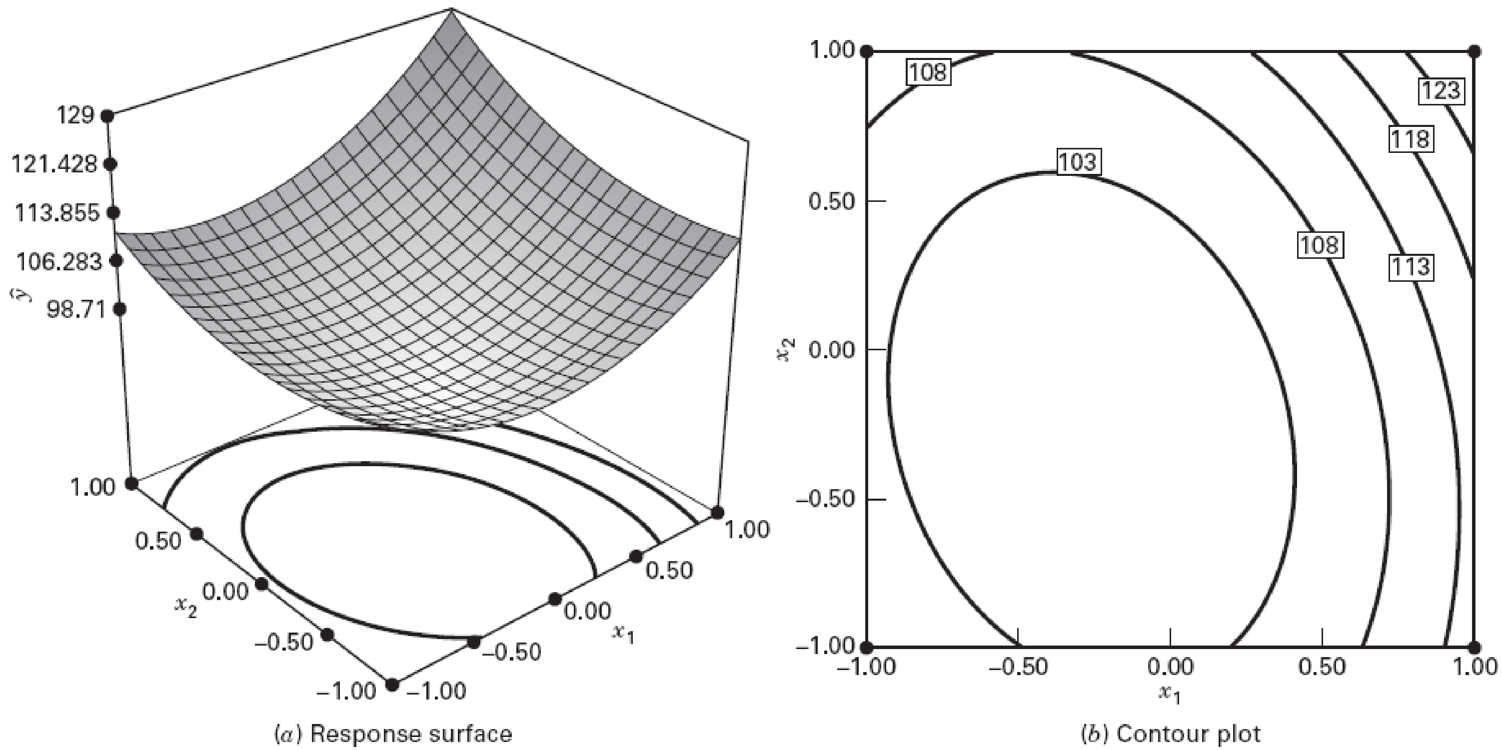
- This point is called the stationary point.
- Stationary point represents…
  - Maximum Point
  - Minimum Point
  - Saddle Point

Maximum Response



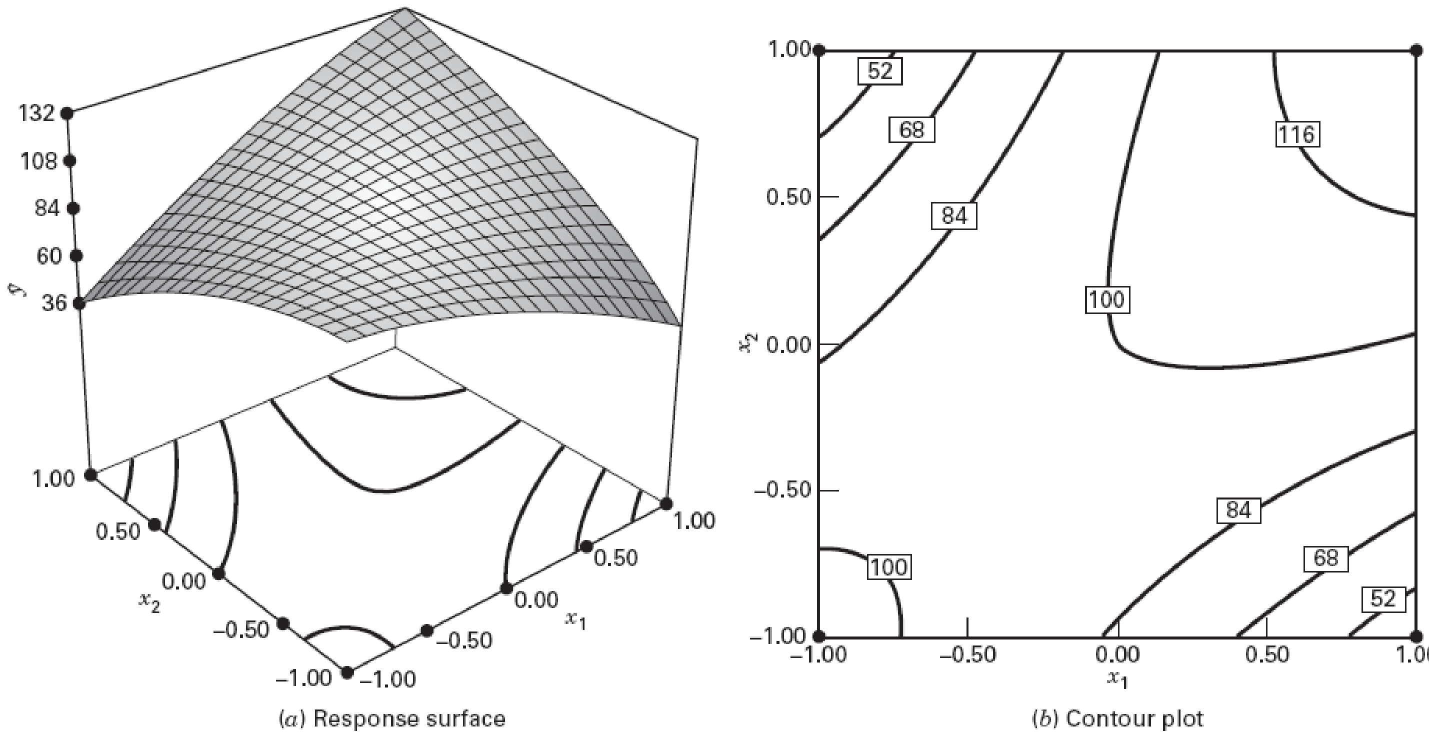
■ **FIGURE 11.6** **Response surface and contour plot illustrating a surface with a maximum**

Minimum Response



■ **FIGURE 11.7 Response surface and contour plot illustrating a surface with a minimum**

Saddle Point



■ **FIGURE 11.8 Response surface and contour plot illustrating a saddle point (or minimax)**

Finding the Stationary Point

After fitting a second order model

$$\hat{y} = \hat{\beta}_0 + \sum \hat{\beta}_i \, x_i + \sum \hat{\beta}_{ii} \, x_i^2 + \sum \sum \hat{\beta}_{ij} \, x_i x_j$$

take the partial derivatives with respect to the x_i's and set to zero

$$\partial \hat{y} / \partial x_1 = \partial \hat{y} / \partial x_2 = \cdots = \partial \hat{y} / \partial x_k = 0$$

Finding the Stationary Point

Matrix view

$$\hat{y} = \hat{\beta}_0 + \mathbf{x}'\mathbf{b} + \mathbf{x}'\mathbf{B}\mathbf{x}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_k \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} \hat{\beta}_{11}, & \hat{\beta}_{12}/2, & \ldots, & \hat{\beta}_{1k}/2 \\ & \beta_{22}, & \ldots, & \hat{\beta}_{2k}/2 \\ & & \ddots & \\ \text{sym.} & & & \hat{\beta}_{kk} \end{bmatrix}$$

$$\frac{\partial \hat{y}}{\partial \mathbf{x}} = \mathbf{b} + 2\mathbf{B}\mathbf{x} = \mathbf{0}$$

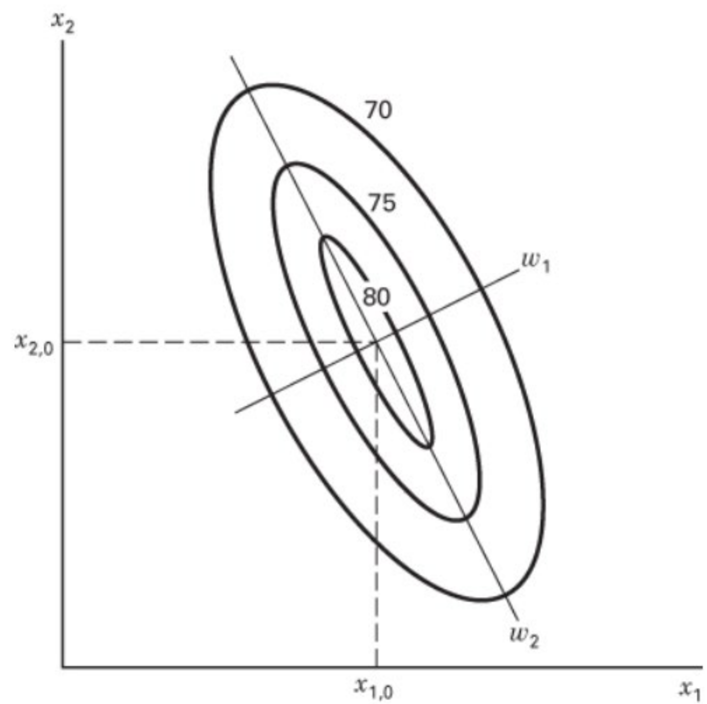
$$\mathbf{x}_s = -\frac{1}{2}\mathbf{B}^{-1}\mathbf{b}$$

Characterization of the Response Surface

- After we have found the stationary point
- Find what type of surface we have
  - Graphical Analysis: contour plot
  - Canonical Analysis
- Determine the sensitivity of the response variable to the optimum value
  - Canonical Analysis

Canonical Analysis

- Transform the model into a new coordinate system
- Origin at the stationary point
- Rotate the axes of this system until they are parallel to the principal axes of the fitted response surface.
- The model in the new system is



$$\hat{y} = \hat{y}_s + \lambda_1 w_1^2 + \lambda_2 w_2^2 + \cdots + \lambda_k w_k^2$$

the $\{\lambda_i\}$ are just the **eigenvalues** of the matrix $\mathbf{B}$

$$\mathbf{B} = \begin{bmatrix} \hat{\beta}_{11}, & \hat{\beta}_{12}/2, & \ldots, & \hat{\beta}_{1k}/2 \\ & \beta_{22}, & \ldots, & \hat{\beta}_{2k}/2 \\ & & \ddots & \\ \text{sym.} & & & \hat{\beta}_{kk} \end{bmatrix}$$

Steps of Canonical Analysis:

1. From the model

$$\hat{y} = \hat{\beta}_0 + \sum \hat{\beta}_i x_i + \sum \hat{\beta}_{ii} x_i^2 + \sum \sum \hat{\beta}_{ij} x_i x_j$$

   we have matrix B.

2. Obtain the stationary point x_s.

3. Compute the eigenvalues of B, denoted as $\{\lambda_i\}$, we have the canonical model

$$\hat{y} = \hat{y}_s + \lambda_1 w_1^2 + \lambda_2 w_2^2 + \cdots + \lambda_k w_k^2$$

4. The nature of the response can be determined by the signs and magnitudes of the eigenvalues $\{\lambda_i\}$

  – all positive: a minimum is found
  – all negative: a maximum is found
  – mixed: a saddle point is found

- The response surface is steepest (most sensitive) in the direction (canonical) corresponding to the largest absolute eigenvalue

Return to the example:

- Obtain stationary point

$$\mathbf{b} = \begin{bmatrix} 0.995 \\ 0.515 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} -1.376 & 0.1250 \\ 0.1250 & -1.001 \end{bmatrix}$$

$$\begin{aligned} \mathbf{x}_s &= -\tfrac{1}{2}\mathbf{B}^{-1}\mathbf{b} \\ &= -\tfrac{1}{2}\begin{bmatrix} -0.7345 & -0.0917 \\ -0.0917 & -1.0096 \end{bmatrix}\begin{bmatrix} 0.995 \\ 0.515 \end{bmatrix} = \begin{bmatrix} 0.389 \\ 0.306 \end{bmatrix} \end{aligned}$$

In terms of the natural variables, the stationary point is

$$0.389 = \tfrac{\xi_1 - 85}{5} \qquad 0.306 = \tfrac{\xi_1 - 175}{5}$$

$$\xi_1 = 86.95 \qquad \xi_2 = 176.53$$

- Compute eigenvalues of B

$$|\mathbf{B} - \lambda \mathbf{I}| = 0$$

$$\begin{vmatrix} -1.376 - \lambda & 0.1250 \\ 0.1250 & -1.001 - \lambda \end{vmatrix} = 0$$

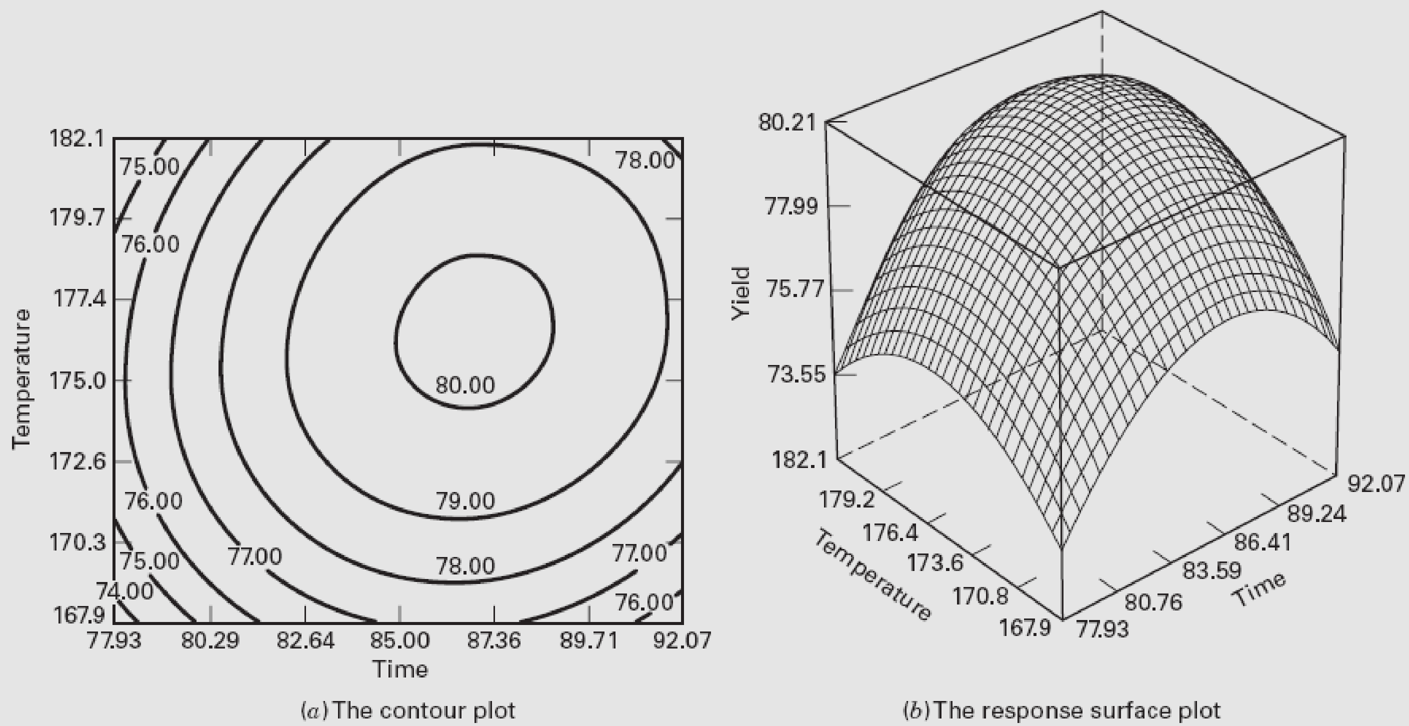
$$\lambda^2 + 2.3788\lambda + 1.3639 = 0$$

$$\lambda_1 = -0.9634 \text{ and } \lambda_2 = -1.4141$$

Thus, the canonical form of the fitted model is

$$\hat{y} = 80.21 - 0.9634 w_1^2 - 1.4141 w_2^2$$

Because both *λ1* and *λ2* are negative and the stationary point is within the region of exploration, we conclude that the stationary point is a maximum.



■ **FIGURE 11.11** **Contour and response surface plots of the yield response, Example 11.2**